

Effect of Electroacupuncture Based on ERAS for Preoperative Anxiety in Breast Cancer Surgery: A Single-Center, Randomized, Controlled Trial

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Abstract

• The effect of Electroacupuncture (EA) based on ERAS on preoperative anxiety in patients undergoing conservative breast surgery was evaluated. • 144 female were grouped into conventional, preoperative EA, intraoperative EA, and combination of preoperative and intraoperative EA. • Preoperative EA alleviated anxiety in the preoperative waiting area, and improved sleep quality. Combined techniques may more effectively improve postoperative quality of life.

Background: To evaluate the effect of electroacupuncture (EA) based on enhanced recovery after surgery on preoperative anxiety in patients undergoing breast cancer surgery. **Patients and Methods:** This was a single-center, randomized, controlled, single-blind clinical trial. Between December 2018 and 2019, 144 female undergoing breast conserving surgery were assigned to conventional (A), preoperative EA (B), intraoperative EA (C), and combination of preoperative and intraoperative EA (D) groups. Primary outcome was the self-rating anxiety scale. Secondary outcomes included visual analogue scale, quality of recovery 40, postoperative complications, and acupuncture-related adverse reactions. **Results:** 141 patients completed the trial. Groups B and D self-rating anxiety scale were significantly lower than A and C ($P < .01$); the sleep quality was significantly better ($P < .01$). The incidence of nausea at 6 hours postoperatively was significantly lower in group D than other groups ($P < .007$); the incidence of vomiting at 6 hours postoperatively was better than group A ($P < .007$). visual analogue scale at 24 hours postoperatively was significantly different between groups A, B, and D ($P < .01$). Quality of recovery 40 total score at 24 hours postoperatively in group D was significantly higher than A and B ($P < .05$). The 72-hour postoperative emotional state was most significantly improved in group D ($P < .05$), while groups B and C showed interactive effect ($P < .05$). **Conclusion:** Preoperative EA alleviated anxiety in the preoperative waiting area, and improved sleep quality. Combination of preoperative and intraoperative EA may be more effective in improving postoperative quality of life. **Trial registration:** Chinese Clinical Trial Registry, ChiCTR1800019979. Registered on December 10, 2018. (<http://www.chictr.org.cn/edit.aspx?pid=27653&htm=4>).

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Introduction

Breast cancer is the most prevalent cancer worldwide. An estimated 7.8 million women were diagnosed with breast cancer in the past 5 years.¹ Surgical resection offers the best chance of cure for early-stage breast cancer and therefore represents the main curative treatment.² However, preoperative anxiety induced by surgery is associated with complications which can lead to delayed recovery, poorer health-related quality of life, and greater hospital costs.^{3–5}

Abbreviations: ERAS, Enhanced recovery after surgery; SAS, Self-rating anxiety scale; VAS, Visual Analogue Scale; TCM, Traditional Chinese medicine; PONV, Postoperative nausea and vomiting.

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More importantly, patients with breast cancer may have a higher risk of preoperative anxiety.⁶ Administration of anxiolytic medication and psychological nursing have been the traditional methods for reducing preoperative anxiety.⁷ However, these drugs often cause serious side effects. A higher risk of postoperative opioid use is a common complication which tends to affect outcomes,⁸ while psychological nursing also has certain limitations. Efforts have been made to identify more effective and safe therapies.

Enhanced recovery after surgery (ERAS) pathways were initially proposed by a Danish scholar in 1995 who envisaged use of evidence-based multi-modal interventions in the perioperative period to alleviate or improve the psychological or physiological stress response of patients.^{9,10} Studies have shown that ERAS protocols for perioperative care can lead to improvement in clinical outcomes and cost savings,^{11,12} and especially ameliorate preoperative anxiety.¹³

Acupuncture therapy as an ancient Chinese therapeutic modality has been used in ERAS pathways with the aim to reduce perioperative pain and postoperative nausea and vomiting.^{14,15} Although

a few studies have shown that acupuncture therapy can reduce preoperative anxiety,¹⁶ these studies were mainly conducted in the context of neuro-, cataract, and gynecological surgeries; in addition, the timing of acupuncture intervention in these studies was not uniform.^{7,17,18} There is a paucity of evidence regarding acupuncture in breast cancer surgery, particularly in terms of the timing of acupuncture which will produce better effects.

In this study, we evaluated the use of electroacupuncture (EA) to alleviate preoperative anxiety and improve postoperative quality of life in patients undergoing breast cancer surgery and compared the benefits of acupuncture performed at different time-points (1 day before surgery versus on the day of surgery versus combination of 1 day before surgery plus on the day of surgery). We hypothesized that the different timing of acupuncture application has differential benefits for enhanced recovery in breast cancer surgery.

Material and Methods

Ethical Approval

The study was approved by the Ethics Committee of Shuguang Hospital (approval no. 2018-624-53-01) and conducted in accordance with the 1964 Helsinki Declaration or comparable standards. Written informed consent was obtained from all participants prior to their enrolment. The trial registration number: ChiCTR1800019979. The full protocol can be accessed online at <http://www.chictr.org.cn/edit.aspx?pid=27653&htm=4>.

Clinical Data

A total of 144 patients who were admitted for breast conserving surgery between December 2018 and December 2019 were recruited. The operation was completed by the same group of surgeons using standardized operating procedures. The flow chart of study is shown in Figure 1.

Inclusion criteria: (1) cases that conformed with the diagnostic criteria in the Guidelines for the diagnosis and treatment of breast cancer (2011 edition) for TCM and Western medicine¹⁹; TNM clinical stage I-III, and no signs of systemic metastasis; (2) age range: 16-70 years; (3) American Association of Anesthesiologists (ASA) grade I-II¹⁵; (4) no history of immune disorders; (5) provision of written informed consent.

Exclusion criteria: (1) patients with malignant hypertension, diabetes, severe heart disease, or implanted pacemaker; (2) symptoms of insomnia or history of intake of sleeping pills in the preceding six months; (3) past history of anxiety or depression; (4) coagulation dysfunction or allergies; (5) pregnant or lactating women; (6) patients with mental illness or dementia; (7) patients with local skin infection at the acupoints; (8) patients participating in other clinical trials within 4 weeks.

Suspension/withdrawal/dropout criteria: (1) occurrence of serious adverse events; (2) incomplete operation or postoperative pathological results not consistent with breast cancer; (3) change of experimental plan due to surgery or anesthesia-related adverse event; (4) incomplete data or inability to evaluate all study outcomes and related side effects; (5) voluntary withdrawal by the patient; (6) inability of the patient to cooperate well or adapt to electroacupuncture; (7) inability of the patient to follow instructions; (8) unmasking of the group identity; (9) lost to follow-up.

Study Design

This was a single-center, parallel group, randomized controlled trial. A random number table was generated using a computer and group allocation was performed using sealed opaque envelope. The assessors were blinded to the group identity.

Experimental Grouping

A total of 144 patients were randomly assigned to four groups ($n = 36$ per group) as follows: conventional (A); preoperative acupuncture (B); intraoperative acupuncture (C); and combined technique acupuncture (D) groups. All patients were operated by the same group of surgeons and anesthesiologists. Group A did not receive acupuncture intervention, group B received EA treatment 1 day prior to surgery, group C received EA treatment on the day of surgery, while group D received treatment as composite techniques of groups B and C.

Experimental Procedure

Group A.

One Day Before Operation. Baseline data were collected 1 day before the operation including body mass index (BMI), heart rate, blood pressure, preoperative self-rating anxiety scale (SAS) score, and the quality of recovery-40 (QoR-40) scale. Written informed consent was obtained. Sleep monitoring was performed at 21:00 h with mega view sleep monitor detection (ZG-S01A, Shanghai MegaHealth Information Technology Co. Ltd., Shanghai, P.R. China).

Day of Operation. In the preoperative waiting area, the patient was advised to fill in the SAS form. Electrocardiogram (ECG) monitoring was performed. In the operating room, an intravenous line was established and the patient was administered Ringer's lactate solution. Blood pressure (mean arterial pressure), ECG, and pulse oxygen saturation (SpO_2) were monitored. Total intravenous anesthesia was used for tracheal intubation. Propofol 1-2 mg/kg, sufentanil 0.3 μ g/kg, and rocuronium 0.6 mg/kg were used for anesthesia induction. During operation, propofol 2-3.5 μ g/mL and remifentanyl 0.05-0.2 μ g/kg/min were administered using target-controlled injection pump. Sufentanil dose was calibrated as per need, and 1 of 5 of the initial dose of rocuronium was added 30-40 minutes after the operation. Drug dose was adjusted to achieve the bispectral index (BIS) in the range 40-60 without body movement response to stimuli. The dosage of remifentanyl and sufentanil were adjusted in case of marked changes in heart rate and blood pressure during skin incision. If BIS did not reach the target value, the propofol concentration was adjusted until the BIS value reached the target.

Group B.

One Day Before Operation. Baseline data collection and procedures were conducted as in group A. In addition, acupuncture intervention was performed at Shenmen (HT 7), Neiguan (PC 6), Yintang (EX-HN3), and Baihui (DU 20) acupoints using reinforcing-reducing method after needle insertion. Sterilized disposable needles were inserted with both hands under aseptic

Figure 1 Flow chart of study.

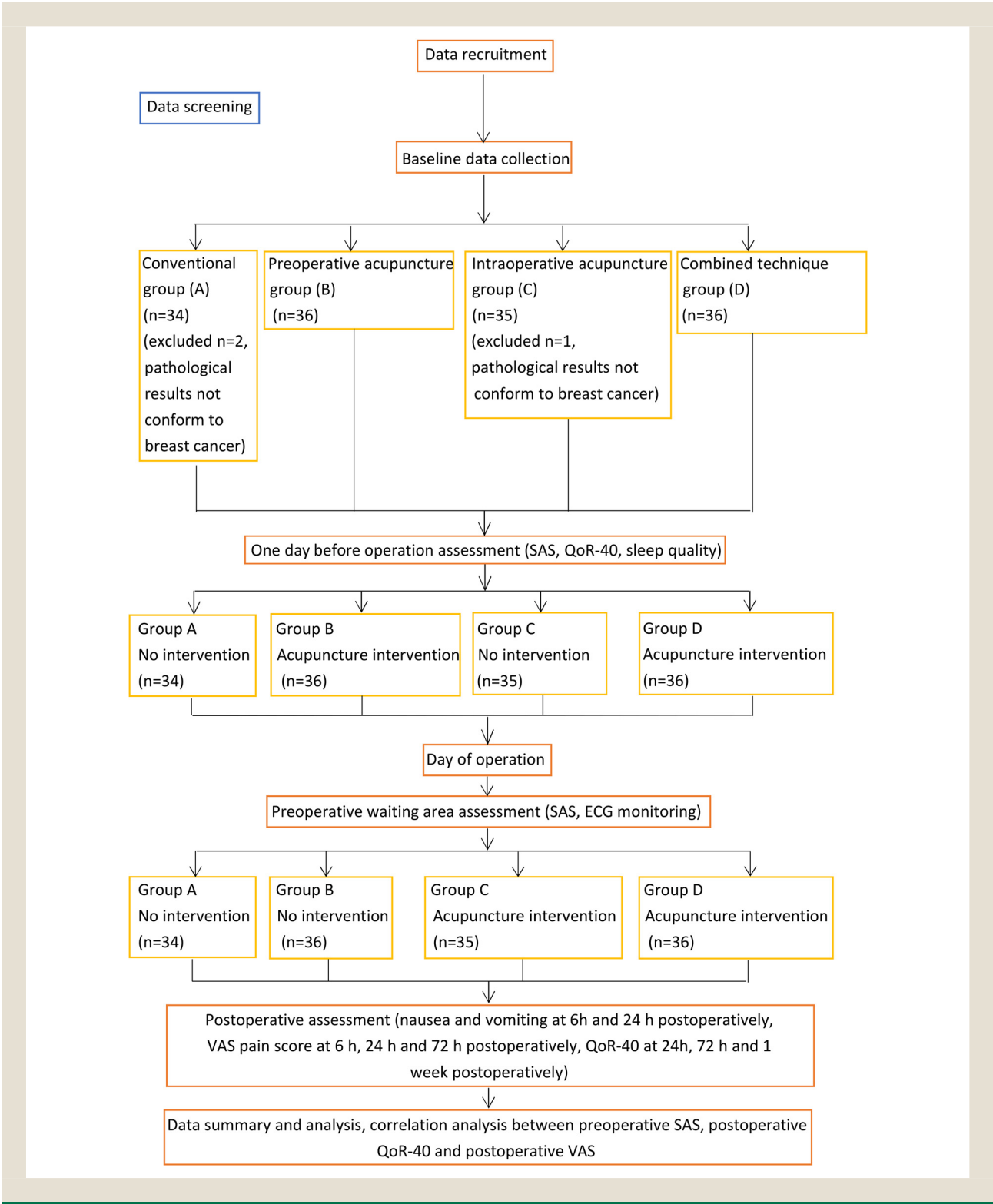


Table 1 Comparison of Baseline Characteristics of Patients in Each Study Group.

Variables	Conventional (group A, n = 34)	Preoperative acupuncture (group B, n = 36)	Intraoperative acupuncture (group C, n = 35)	Combined Technique acupuncture (group D, n = 36)	F	P
Age (years)	53.32 ± 10.78	51.92 ± 10.69	53.51 ± 10.85	52.14 ± 12.11	0.19	.90
Operation duration (min)	130.12 ± 10.04	128.36 ± 6.76	128.43 ± 8.15	126.61 ± 8.29	1.03	.38
BMI	20.93 ± 2.01	21.71 ± 2.49	22.06 ± 2.42	21.44 ± 1.76	1.64	.18
Pre-operative blood glucose (mmol)	5.52 ± 0.98	5.30 ± 0.96	5.01 ± 0.73	5.49 ± 1.14	2.14	.10
SAS score (point)	46.06 ± 5.08	47.92 ± 4.05	46.00 ± 2.58	47.78 ± 4.92	2.14	.10
QoR-40 score (point)	183.09 ± 6.21	183.47 ± 6.80	183.03 ± 5.80	184.86 ± 6.91	0.63	.60
Heart rate (bpm)	80.94 ± 4.31	82.39 ± 3.94	82.37 ± 4.39	81.31 ± 4.55	1.04	.38
Systolic blood pressure (mmHg)	129.53 ± 6.03	129.47 ± 5.86	128.49 ± 5.63	128.92 ± 6.02	0.25	.86
Diastolic blood pressure (mmHg)	77.41 ± 4.02	77.36 ± 3.97	77.37 ± 4.85	75.64 ± 4.25	1.48	.22

Data presented as mean ± standard deviation.

conditions. Electroacupuncture was connected to Yintang-Baihui and Shenmen-Neiguan. The G6805-2 therapy device was used and the frequency was adjusted to 2 Hz, continuous wave type, with current intensity set according to tolerance of the patient, continuously for 30 minutes. The acupoints were selected in accordance with the recommended guideline²⁰ and national standard of the People's Republic of China GB 12346-90 Location of points.

Day of operation

Data collection and procedure were conducted as in group A.

Group C.

One Day Before Operation. The baseline data collection and procedure were conducted as in group A.

Day of Operation. While waiting in the preoperative waiting area, the patient was advised to fill in the SAS form. ECG monitoring was performed, and acupuncture intervention was conducted at bilateral Hegu (LI 4), Neiguan (PC 6), and Liangqiu (ST 34) using reinforcing-reducing method after needle insertion. Sterilized disposable needles were inserted with both hands under aseptic conditions. Electroacupuncture was connected to bilateral Hegu, Neiguan, and Liangqiu (1 inch outwards). The G6805-2 therapy device was used with the same parameters and duration as in group B. In addition, Yintang and Baihui acupoints were used to enhance the sedative effect, and continued for more than 30 minutes. After entering the operating room, the electroacupuncture intervention was continued, and the subsequent steps were the same as those in group A. After completion of the operation, extubation was performed, and the needles were removed after the patient was awake.

Group D.

One Day Before Operation. Baseline data collection and procedure were conducted as in group A. Then, the acupuncture and electroacupuncture intervention was conducted as in group B.

Day of Operation. The data collection and procedure were conducted as in group A, followed by acupuncture and electroacupuncture intervention as in group C.

Endpoints

The primary endpoint was SAS score. The secondary endpoints were sleep quality, postoperative complications (nausea and vomiting, and dizziness at 6 hours and 24 hours postoperatively), VAS pain score, and QOR-40 score (24 hours, 72 hours and 1 week postoperatively), and functional assessment of cancer therapy-breast (1 week and 3 months [postoperative telephonic follow-up]).

Clinical Safety Evaluation

During the course of the study, clinical safety assessment was based on the patient's self-report or daily observation by physician and during follow-up. Any adverse events related to acupuncture were managed as per the guidelines.²¹

Data Analysis

Data analyses were performed using SPSS 19.0 (SPSS Inc., Chicago, IL). Continuous variables are presented as mean ± standard deviation. Differences between two groups were assessed using the *t* test. Analysis of variance and post hoc least significance difference and Student-Newman-Keuls tests were used for multi-group comparisons. Categorical variables are presented as frequency (percentage) and between-group difference assessed using Chi-squared test. Spearman correlation analysis was performed to assess the correlation between pre-operative SAS, postoperative QoR-40, and VAS scores. Statistical significance was defined as *P* < .05.

Sample Size Calculation

This study mainly compared the difference in therapeutic effect between the acupuncture groups and the conventional group. Preoperative SAS was used as the main indicator. Owing to the lack of relevant literature on preoperative anxiety in the use of acupuncture in breast cancer, a preliminary study was conducted to explore

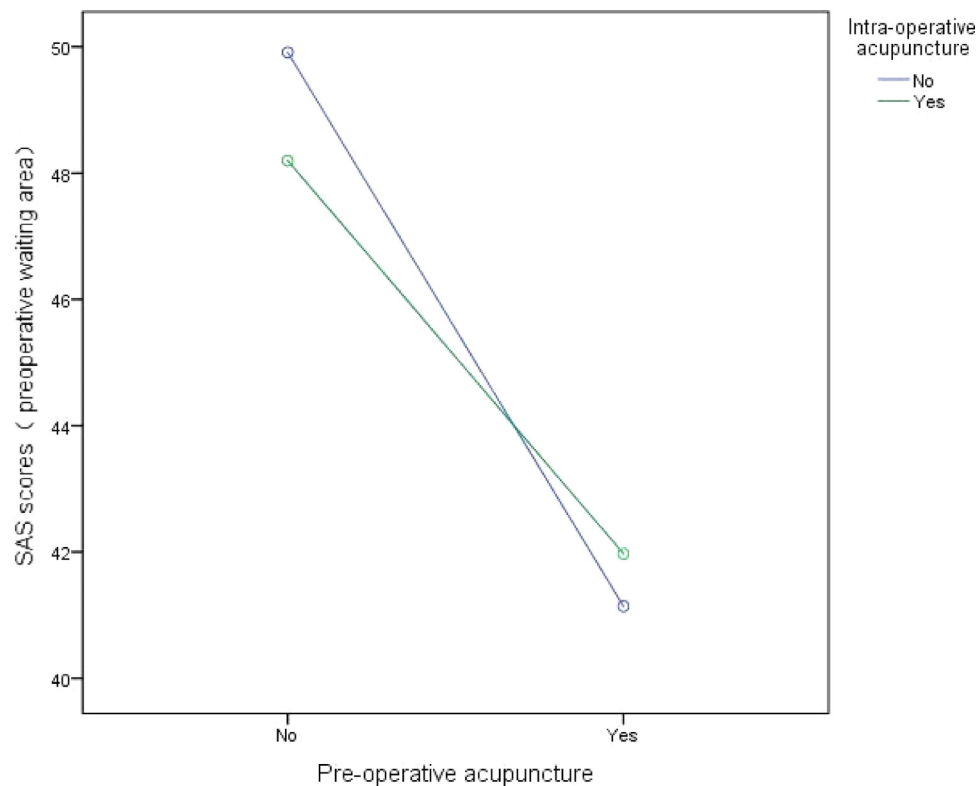
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Table 2 Comparison of SAS Score 1 d Before Operation in the Preoperative Waiting Area in Each Group

Group	<i>n</i>	SAS (1 d Before Operation)	SAS (preoperative Waiting Area)	<i>t</i> (Within Group)	<i>P</i> (Within Group)
Conventional (group A)	34	46.06 ± 5.08	49.91 ± 5.27	−7.35	<.01
Preoperative acupuncture (group B)	36	47.92 ± 4.05	41.14 ± 4.77 ^a	9.41	<.01
Intraoperative acupuncture (group C)	35	46.00 ± 2.58	48.20 ± 3.80	−4.13	<.01
Combined technique acupuncture (group D)	36	47.78 ± 4.92	41.97 ± 4.10 ^a	9.09	<.01
<i>F</i>		2.14	33.35		
<i>P</i>		.10	<.01		

Data presented as mean ± standard deviation.
^a *P* < .01 vs. group A and group C.

Figure 2 Factorial analysis of variance of SAS scores (preoperative waiting area). SAS = self-rating anxiety scale.



the feasibility and to determine the sample size. There were 10 cases in the acupuncture group and 9 cases in the general anesthesia group. The SAS scores of the acupuncture and general anesthesia groups were 43.93 ± 4.98 and 49.63 ± 6.08 , respectively. With the test level $\alpha = 0.05$, power $1 - \beta = 0.90$, and ratio of the 2 groups as 1:1, the calculated sample size²² was 32.26 patients per group. Incorporating a 10% dropout factor, the final sample size was set at 36 patients per group.

Results

Baseline Data Comparison

There were no significant differences between the 4 groups with respect to age, BMI, operation time, preoperative blood glucose level, preoperative anxiety score (SAS), preoperative QOR-40 score, heart rate, or systolic and diastolic blood pressure ($P = .90$, $P = .18$, $P = .38$, $P = .10$, $P = .10$, $P = .60$, $P = .38$, $P = .86$, $P = .22$; respectively) (Table 1). The mean SAS score for groups A, B, C, and

D was 46.06 ± 5.08 , 47.92 ± 4.05 , 46.00 ± 2.58 , and 47.78 ± 4.92 , respectively.

Among all patients recruited between December 2018 and December 2019, 3 dropped out of the study (2 in group A and 1 in group C) as their intraoperative or postoperative pathological results did not conform to breast cancer.

The Primary Outcome

Comparison of SAS before operation. The SAS scores in the preoperative waiting area were significantly different in all four groups ($P < .01$); the scores in groups A (46.06 ± 5.08 vs. 49.91 ± 5.27) and C (46.00 ± 2.58 vs. 48.20 ± 3.80) were significantly increased, while those in groups B (47.92 ± 4.05 vs. 41.14 ± 4.77) and D (47.78 ± 4.92 vs. 41.97 ± 4.10) were significantly decreased compared with 1 day before operation ($P < .01$). There was significant differences between the 4 groups ($P < .01$) (Table 2).

There was a significant difference in the mean SAS scores between preoperative acupuncture intervention and before intervention in the preoperative waiting area ($P < .01$). Preoperative acupuncture was found to effectively reduce the anxiety score of patients in the preoperative waiting area (Figure 2).

The SAS scores in groups B (41.14 ± 4.77) and D (41.97 ± 4.10) were significantly lower than those in groups A (49.91 ± 5.27) and C (48.20 ± 3.80) ($P < .01$). Subsequently, analysis to observe possible interaction between the preoperative and intraoperative interventions, showed that there was a significant difference in the SAS in group B ($P < .01$), but no significant differences in group C and group D ($P > .05$), indicating no interaction between the preoperative and intraoperative acupuncture groups.

Secondary Outcomes

Comparison of Sleep Quality 1 Day Before Operation. The total recorded sleep time was not significantly different among the four groups ($P = .72$) (Table 3); however, the total sleep time, recovery time, light sleep period, and sleep efficiency were significantly different ($P < .01$, all). Groups B (54.89 ± 7.52 minutes) and D (52.11 ± 7.68 minutes) had significantly shorter recovery time than groups A (68.59 ± 7.15 minutes) and C (67.26 ± 7.68 minutes) ($P < .01$). The sleep efficiency in groups B ($87.06 \pm 2.22\%$) and D ($87.34 \pm 1.88\%$) was significantly greater than that in groups A ($83.27 \pm 1.52\%$) and C ($83.84 \pm 1.72\%$) ($P < .01$). The light sleep period and total sleep time in group B (211.5 ± 20.39 minutes and 365.5 ± 37.87 minutes, respectively) and D (203.08 ± 16.33 minutes and 360.42 ± 31.55 minutes, respectively) were significantly longer than those in group A (191.44 ± 23.17 minutes and 341.21 ± 38.43 minutes, respectively) and C (198.97 ± 23.24 minutes and 349.31 ± 37.84 minutes, respectively) ($P < .01$ and $P < .05$ for groups B and D, respectively). This showed that the sleep quality in groups B and D was better than that in the other 2 groups.

Incidence of Nausea and Vomiting at 6 Hours And 24 Hours Postoperatively. The incidence of nausea at 6 hours and 24 hours postoperatively was significantly different among the 4 groups (Chi-squared; $P = .02$ and $P = .01$, respectively) (Table 4). The Bonferroni method was used to correct α for multiple comparisons, and the results showed significant differences when P was less than .007.

Table 3 Comparison of Sleep Quality 1 d Before Operation in the Preoperative Waiting Area in Each Group.

Group	n	Total Recording Time (min)	Total Sleep Time (min)	Recovery Time (min)	Light Sleep Period (min)	Deep Sleep Period (min)	Eye Move-ment Period (min)	Sleep Efficiency (%)
Conventional (group A)	34	409.50 $\pm$ 42.29	341.21 $\pm$ 38.43	68.59 $\pm$ 7.15	191.44 $\pm$ 23.17	95.38 $\pm$ 15.60	54.38 $\pm$ 11.47	83.27 $\pm$ 1.52
Preoperative acupuncture (group B)	36	419.58 $\pm$ 39.02	365.5 $\pm$ 37.87 ^a	54.89 $\pm$ 7.52 ^a	211.5 $\pm$ 20.39 ^a	102.14 $\pm$ 15.15	51.89 $\pm$ 10.34	87.06 $\pm$ 2.22 ^a
Intraoperative acupuncture (group C)	35	416.49 $\pm$ 41.38	349.31 $\pm$ 37.84	67.26 $\pm$ 7.68	198.97 $\pm$ 23.24	98.03 $\pm$ 17.04	52.31 $\pm$ 8.21	83.84 $\pm$ 1.72
Combined technique acupuncture (group D)	36	412.53 $\pm$ 34.85	360.42 $\pm$ 31.55 ^a	52.11 $\pm$ 7.68 ^a	203.08 $\pm$ 16.33 ^a	101.97 $\pm$ 13.66	55.36 $\pm$ 9.57	87.34 $\pm$ 1.88 ^a
F		0.44	3.15	44.25	5.6	1.58	0.99	45.73
P		.72	.03	<.01	<.01	.20	.40	<.01

Data presented as mean $\pm$ standard deviation.

^a $P < .01$ vs. group A and group C.

Table 4 Comparison of the Incidence of Nausea and Vomiting in Each Group 6 h and 24 h Postoperatively.

Group	N	Nausea						Vomiting					
		6 h postoperatively			24 h postoperatively			6 h postoperatively			24 h postoperatively		
		No	Yes	χ^2	P	No	Yes	No	Yes	χ^2	P	No	Yes
Conventional (group A)	34	11 (32.40)	23 (67.60)	9.84	.02	8 (23.50)	26 (76.50) ^a	18 (52.90)	16 (47.10)	8.30	.04	27 (79.40)	7 (20.60)
Preoperative acupuncture (group B)	36	12 (33.30)	24 (66.70)			21 (58.30)	15 (41.70)	22 (61.10)	14 (38.90)			29 (80.60)	7 (19.40)
Intraoperative acupuncture (group C)	35	13 (37.10)	22 (62.90)			20 (57.10)	15 (42.90)	25 (71.40)	10 (28.60)			33 (94.30)	2 (5.70)
Combined technique acupuncture (group D)	36	23 (63.90)	13 (36.10)			32 (88.90)	4 (11.10)	30 (83.30)	6 (16.70) ^b			34 (94.40)	2 (5.60)
Total	141	59 (41.80)	82 (58.20)			81 (57.40)	60 (42.60)	95 (67.40)	46 (32.60)			123 (87.20)	18 (12.80)

Data presented as n (%).

^a $P < .007$ vs. group B; group C, and group D.

^b $P < .007$ vs. group A.

For incidence of nausea at 6 hours postoperatively, pairwise comparisons showed that group D (36.10%) was significantly better than the other three groups (67.60%, 66.70%, and 62.90% for groups A, B, and C, respectively; $P < .007$). For incidence of nausea at 24 hours postoperatively, pairwise comparisons showed significant differences between group A (76.50%) and groups D (11.10%), B (41.70%), and C (42.90%) ($P < .007$).

The incidence of vomiting at 6 hours postoperatively was significantly different among the 4 groups ($P = .04$) (Table 4). Pairwise comparisons showed significant difference between groups A (47.10%) and D (16.70%) ($P < .007$). The incidence of vomiting at 24 hours postoperatively was not significantly different among the four groups ($P = .90$).

Comparison of VAS Pain Score at 6 Hours, 24 Hours, and 72 Hours Postoperatively. The results of Kruskal-Wallis H rank-sum test demonstrated no significant differences in the VAS pain scores at 6 hours and 72 hours postoperatively among the 4 groups ($P = .08$, $P = .14$, respectively), but there was a significant difference at 24 hours postoperatively ($P < .01$) (Table 5). Kruskal-Wallis H pairwise comparisons showed significant differences in the VAS pain scores at 24 hours postoperatively between groups A, B, and D ($P < .01$). There was significant difference between group C and group A ($P < .05$).

Comparison of QoR-40 Scores at 24 Hours, 72 Hours, and 1 Week Postoperatively. There were no significant differences in the preoperative QoR-40 scores among the 4 groups ($P = .60$) (Table 6), however, there were significant differences in the QoR-40 total score at 24 hours postoperatively ($P = .03$). The QoR-40 total score at 24 hours postoperatively in group D (156.08 ± 3.72) was significantly higher than that in groups A (153.50 ± 5.31) and B (152.67 ± 5.64) ($P < .05$).

This showed that intraoperative acupuncture improved the postoperative quality of life score, and there was no interaction between the preoperative and intraoperative acupuncture groups.

There were significant differences in the QoR-40 total score at 72 hours postoperatively among the four groups ($P < .01$) (Table 6). The emotional state score in group D (37.53 ± 2.21) was significantly higher than that in group A (36.56 ± 2.35) ($P = .03$), and the physical comfort score (56.83 ± 4.56) was significantly higher than group A (52.41 ± 6.82) ($P < .01$).

For variables which demonstrated significant differences (total score, emotional state, physical comfort), analysis of variance of factorial design was performed. The QoR-40 total score at 72 hours postoperatively was significantly different in group C ($P < .01$), while there were no significant differences in group B and group D ($P > .05$). These findings indicated that intraoperative acupuncture significantly increased the total score and improved the quality of life of patients, and that there was no interaction between the preoperative and intraoperative acupuncture groups. Group D showed a significant effect on the emotional state at 72 hours postoperatively ($P < .05$). There were no significant differences in groups B and C ($P > .05$); however, that there was an interactive effect between them ($P < .05$), and the 2 groups needed to work together to improve this

Table 5 VAS Pain Scores at Different Time-points After Operation

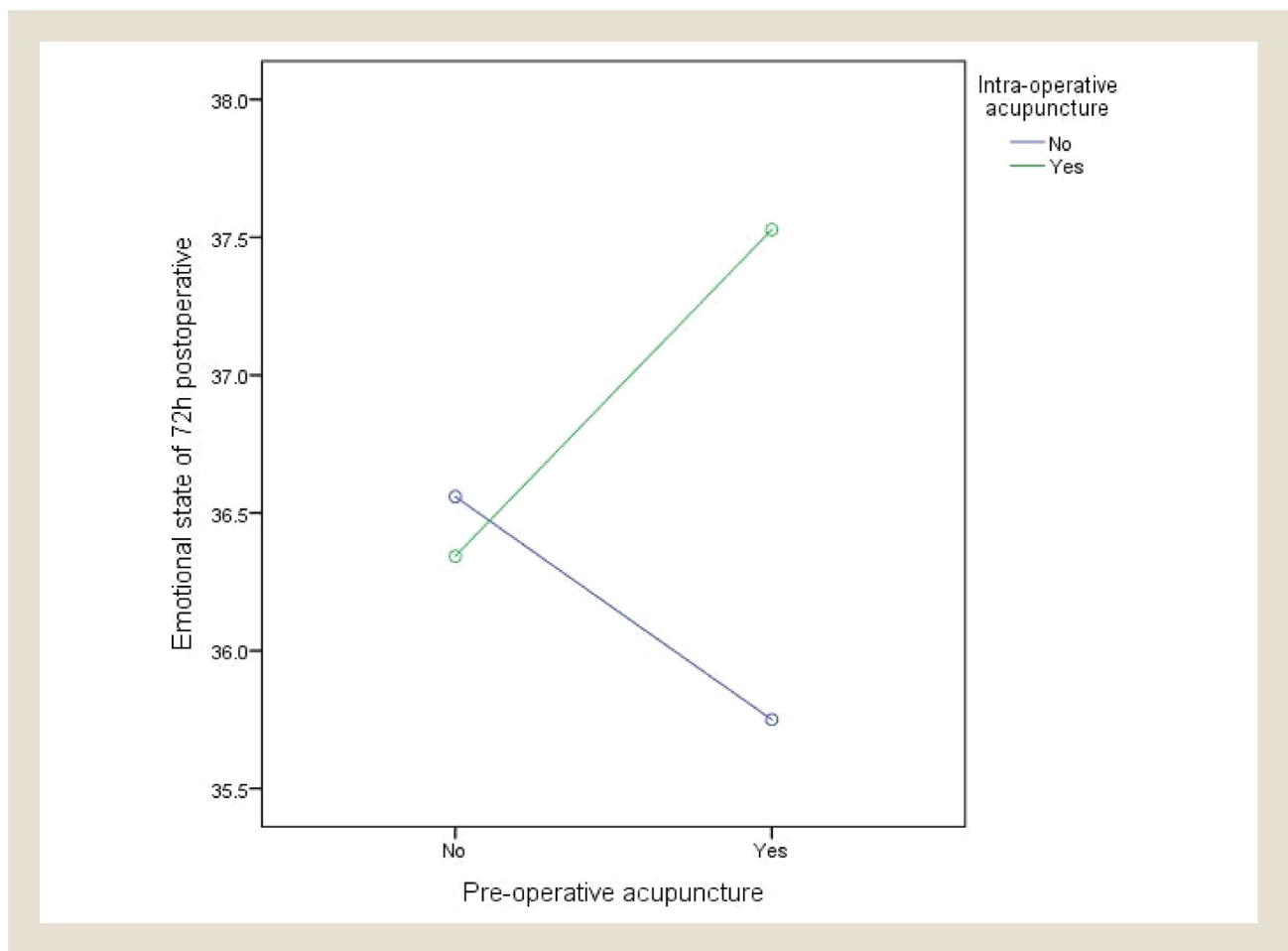
Group	Number of Cases	Postoperative 6 h	Postoperative 24 h	Postoperative 72 h
Conventional (group A)	34	2.00 (1.00, 3.00)	2.00 (1.00, 2.00)	0.00 (0.00, 0.00)
Preoperative acupuncture (group B)	36	2.00 (1.00, 2.75)	2.00 (1.00, 2.00)	0.00 (0.00, 0.00)
Intraoperative acupuncture (group C)	35	2.00 (1.00, 3.00)	1.00 (0.00, 2.00) ^a	0.00 (0.00, 0.00)
Combined technique acupuncture (group D)	36	1.00 (1.00, 2.00)	0.00 (0.00, 1.00) ^b	0.00 (0.00, 0.00)
χ^2		6.84	30.13	5.51
<i>P</i>		.08	<i>P</i> < .01	.14

Analysis was performed by Kruskal-Wallis H rank-sum test (M [P25, P75]).

^a *P* < .05 vs. group A;

^b *P* < .01 vs. group A and group B.

Figure 3 Factorial analysis of variance of emotional state scores (72 hours postoperative).



component of the score. The score of patients in group D was the highest (37.53 ± 2.21) (Figure 3).

The physical comfort score at 72 hours postoperatively in group C was significantly different ($P < .01$), but there were no significant differences in groups B and D ($P > .05$). This showed that intraoperative acupuncture can significantly improve the physical comfort score and the quality of life of patients, and that there was no interaction between the preoperative and intraoperative acupuncture groups.

There were significant differences between all 4 groups with respect to the QoR-40 total scores at 1 week postoperatively ($P = .04$) (Table 6). The total score of group D (181.75 ± 6.02) was higher than that in groups A (178.03 ± 6.26) and B (178.72 ± 5.87) ($P < .05$).

For those components which demonstrated significant differences (total score), analysis of variance of factorial design was performed. The QoR-40 total score at 1 week postoperatively in group C (180.69 ± 5.67) was significantly different ($P < .01$), but there was

Table 6 Comparison of Preoperative and 24 h, 72 h, and 1-wk Postoperative QoR-40 Scores in Each Group.

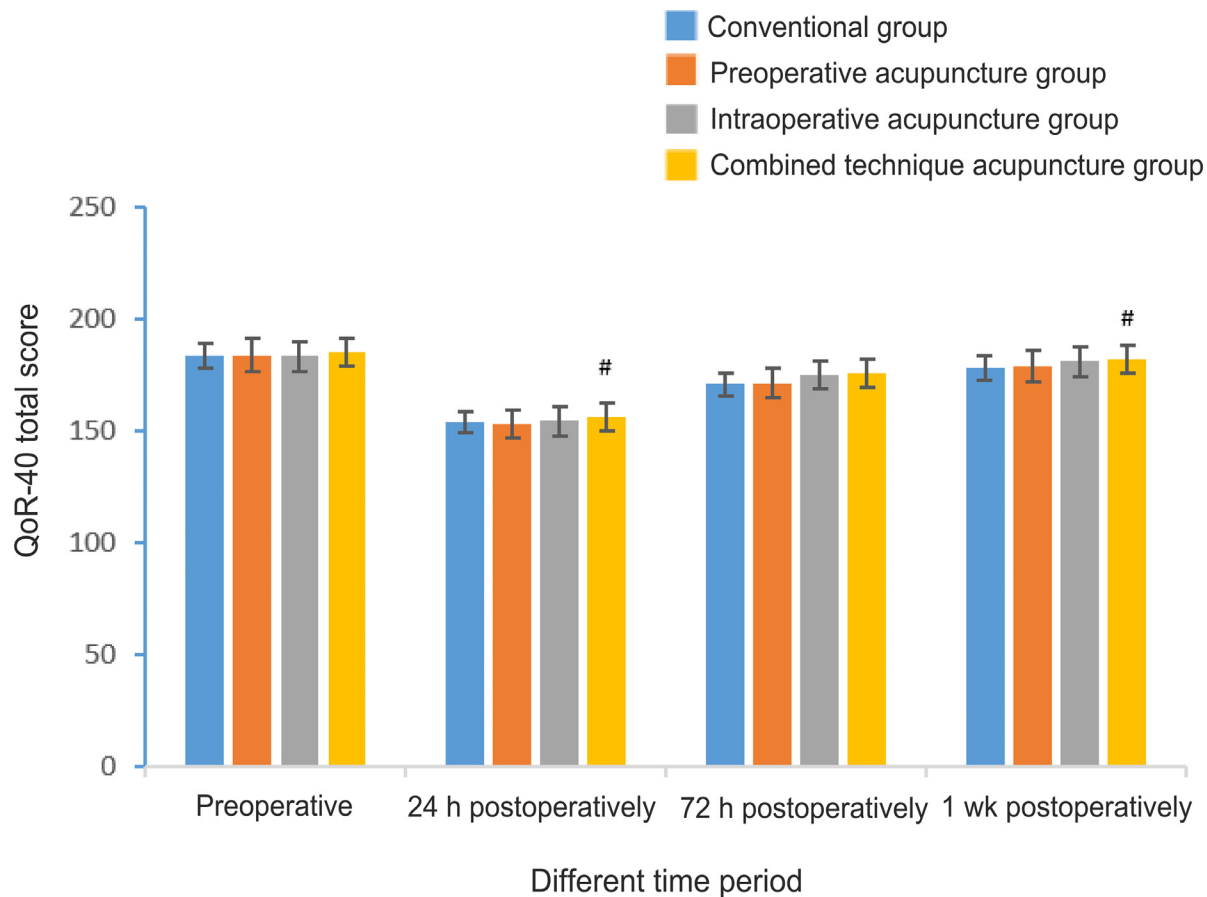
Group			Conventional (Group A)	Preoperative Acupuncture (Group B)	Intraoperative Acupuncture (Group C)	Combined Technique Acupuncture (Group D)	F	P
N			34	36	35	36		
Emotional state	Preoperative		40.62 ± 1.83	40.39 ± 1.81	40.94 ± 1.64	40.33 ± 1.97	0.82	.48
	Postoperative	24 h	34.50 ± 2.95	35.19 ± 2.84	34.91 ± 3.00	35.89 ± 2.78	1.44	.24
		72 h	36.56 ± 2.35	35.75 ± 2.59	36.34 ± 2.71	37.53 ± 2.21 ^{a,b}	3.21	.03
		1 wk	40.35 ± 1.65	40.28 ± 1.73	40.51 ± 1.56	40.83 ± 1.89	0.74	.53
Physical comfort	Preoperative		52.06 ± 6.76	51.78 ± 7.19	51.34 ± 5.50	53.50 ± 6.24	0.75	.53
	Postoperative	24 h	43.03 ± 5.76	41.92 ± 5.48	43.80 ± 5.17	44.58 ± 4.71	1.66	.18
		72 h	52.41 ± 6.82	54.56 ± 5.99	56.54 ± 4.69 ^c	56.83 ± 4.56 ^c	4.68	<.01
		1 wk	48.06 ± 6.00	49.06 ± 5.96	50.46 ± 5.64	50.97 ± 5.58	1.83	.14
Self care ability	Preoperative		23.82 ± 0.67	23.97 ± 0.81	23.80 ± 0.80	23.75 ± 0.73	0.58	.63
	Postoperative	24 h	16.00 ± 1.13	15.26 ± 1.36	15.58 ± 1.13	15.78 ± 1.10	2.46	.07
		72 h	17.68 ± 1.39	17.28 ± 1.49	17.86 ± 1.38	17.50 ± 1.44	1.07	.36
		1 wk	23.53 ± 0.90	23.69 ± 0.79	23.66 ± 0.84	23.53 ± 0.61	0.43	.74
Psychological support	Preoperative		31.85 ± 1.52	32.56 ± 1.50	32.49 ± 1.63	32.50 ± 1.72	1.49	.22
	Postoperative	24 h	31.09 ± 2.01	30.94 ± 2.24	31.14 ± 2.21	30.83 ± 2.29	0.15	.93
		72 h	32.56 ± 1.38	32.47 ± 1.56	32.63 ± 1.52	32.44 ± 1.44	0.12	.95
		1 wk	31.97 ± 1.57	31.67 ± 1.43	32.00 ± 1.61	32.00 ± 1.64	0.39	.76
Pain	Preoperative		34.74 ± 0.57	34.78 ± 0.49	34.46 ± 0.82	34.78 ± 0.59	2.14	.1
	Postoperative	24 h	28.88 ± 1.55	28.83 ± 1.30	29.11 ± 1.49	29.19 ± 1.67	0.48	.7
		72 h	31.38 ± 1.50	31.06 ± 1.19	31.34 ± 1.35	31.31 ± 1.35	0.42	.74
		1 wk	34.12 ± 0.84	34.03 ± 0.88	34.06 ± 0.91	34.42 ± 0.69	1.65	.18
Total score	Preoperative		183.09 ± 6.21	183.47 ± 6.80	183.03 ± 5.80	184.86 ± 6.91	0.63	.60
	Postoperative	24 h	153.50 ± 5.31	152.67 ± 5.64	154.23 ± 5.12	156.08 ± 3.72 ^{a,d}	3.05	.03
		72 h	170.59 ± 7.26	171.11 ± 6.93	174.71 ± 6.93	175.61 ± 7.00	4.56	<.01
		1 wk	178.03 ± 6.26	178.72 ± 5.87	180.69 ± 5.67	181.75 ± 6.02 ^{a,d}	2.94	.04

Data presented as mean ± standard deviation. Analysis was performed with ANOVA, followed by post hoc LSD and SNK.

Abbreviations: ANOVA = analysis of variance; LSD = least significance difference; QoR-40 = quality of recovery-40.

^a $P < .05$.^b $P < .01$ vs. group B.^c $P < .01$ vs. group A.^d $P < .05$.

Figure 4 QoR-40 score changes at each time-point in the four groups. # $P < .05$ vs. group A and group B. QoR-40 = quality of recovery-40.



no significant difference in group B (178.72 ± 5.87) and group D (181.75 ± 6.02) ($P > .05$). This showed that intraoperative acupuncture increased the total score 1 week later and improved the quality of life of patients, and that there was no interaction between the preoperative and intraoperative acupuncture groups. As shown in Figure 4, the QoR-40 total scores of the four groups were the lowest at 24 hours, and gradually recovered to the preoperative state within 1 week after the operation.

Correlation Analysis Between Preoperative SAS and Postoperative QoR-40. Spearman correlation analysis revealed a negative correlation between SAS (preoperative waiting) and 24 hours postoperatively emotional state ($r = -0.205$) (Table 7). In other words, the higher the anxiety score in the preoperative waiting area, the lower was the 24 hours postoperatively QoR-40 emotional state score.

We also observed a negative correlation between SAS (1 day before operation) and emotional state at 72 hour postoperatively ($r = -0.204$), that is, the higher the anxiety score 1 day before the operation, the lower was the QoR-40 emotional state score at 72 hours postoperatively.

SAS (1 day before operation) also showed a negative correlation with 1 week postoperatively physical comfort score ($r = -0.225$) ($P < .01$), VAS score at 1 week after operation ($r = -0.24$) ($P < .01$), and the QoR-40 total score at 1 week postoperatively ($r = -0.254$) ($P < .01$).

SAS score (pre-operative waiting) showed negative correlation with physical comfort at 1 week postoperatively ($r = -0.258$; $P < .01$), VAS score at 1 week postoperatively ($r = -0.225$; $P < .01$) and the QoR-40 total score at 1 week post-operatively ($r = -0.242$; $P < .01$).

Discussion

Preoperative anxiety is one of the main problems experienced by hospitalized surgical patients, especially those with breast cancer, and it may increase intraoperative risk and worsen postoperative prognosis.²³ Studies have shown that acupuncture therapy can significantly reduce preoperative anxiety,^{7,18,24} but there is no consensus on the optimal timing of the intervention before surgery.

In this study, we explored the effect of EA on preoperative anxiety at different time-points. Our results showed that preoperative EA decreased SAS scores of patients compared to conventional therapy.

Table 7 Results of Spearman Correlation Analysis Showing Correlation Between Preoperative SAS and 24 h, 72 h, and 1 wk Postoperative QoR-40 Score.

	Emotional State			Physical Comfort			Self-care Ability			Psychological Support			Pain			Total Score		
	Postoperative			Postoperative			Postoperative			Postoperative			Postoperative			Postoperative		
	24 h	72 h	1 wk	24 h	72 h	1 wk	24 h	72 h	1 wk	24 h	72 h	1 wk	24 h	72 h	1 wk	24 h	72 h	1 wk
SAS (1 d before operation)	0.036	-0.204 ^a	0.085	-0.007	-0.038	-0.225 ^b	0.095	-0.008	-0.138	-0.035	0.003	-0.060	0.053	-0.045	-0.240 ^b	0.036	-0.103	-0.254 ^b
SAS (Preoperative waiting)	-0.205 ^a	-0.138	0.135	0.036	-0.138	-0.258 ^b	0.071	0.054	-0.148	-0.048	-0.002	-0.018	0.032	0.053	-0.225 ^b	-0.060	-0.138	-0.242 ^b

Abbreviation: SAS = self-rating anxiety scale.
^a $P < .05$.
^b $P < .01$ indicates significant correlation between 2 variables.

Moreover, there was no interaction between the preoperative and intraoperative EA groups. At the same time, EA was associated with better sleep quality, less incidence of PONV, and lower VAS pain scores. Factorial analysis showed that preoperative and intraoperative EA had an interactive effect on the quality of life, and that the effect was better.

More than 90% of adults patients undergoing scheduled elective surgery experience preoperative anxiety, which often lasts until the surgery, and the most severe anxiety is experienced in the surgery waiting area.^{25,26} Many studies have reported the intervention duration which varied from several minutes to several days prior to a surgical procedure.^{24,27,28} Therefore, we designed the trial to conduct EA for patients with preoperative anxiety on the day before surgery, intraoperatively, and a combination of preoperative and intraoperative EA. Considering that performing EA in the preoperative waiting area may cause stress response and increase the patient's anxiety, we conduct EA intervention on the day before the operation. Our results demonstrated that preoperative EA can effectively reduce the anxiety score of breast cancer patients. This finding was different from the previous studies.^{17,29} This difference may be attributable to the different timing of intervention and choice of acupuncture points in our study. In the contemporary literature, EX-HN3, PC6, DU20, and HT7 are the most frequently used acupuncture points,^{16,30,31} which are associated with calming and soothing effects. Moreover, these acupuncture points can reduce dependence on drug treatment besides significantly relieving anxiety.³²

Studies have demonstrated a strong correlation between preoperative anxiety and insomnia. A prospective cohort study suggested that individuals with insomnia display more anxiety symptoms.³³⁻³⁵ Pairwise comparison showed that, the preoperative EA group and combination of preoperative and intraoperative EA group had shorter recovery period, greater sleep efficiency, and longer light sleep and total sleep period than the other groups. Similarly, these groups had lower SAS scores which demonstrated that EA could improve sleep quality to mitigate preoperative anxiety.

Postoperative nausea and vomiting (PONV) refer to nausea or vomiting occurring within 24 hours postoperatively.³⁶ In a recent prospective observational study, preoperative anxiety was identified as a predictor of PONV.³⁷ In this study, the combination of preoperative and intraoperative EA group showed the best effect in alleviating postoperative nausea, and 6 hours postoperative vomiting, but there was no significant different among the 4 groups with respect to 24 hours postoperative vomiting. On the one hand, this showed that EA may reduce PONV; on the other hand, this showed that the relationship between preoperative anxiety and PONV was unclear. Coincidentally, a study³⁸ suggested a weak and nonunique association between anxiety and PONV. Therefore, the relationship between preoperative anxiety and PONV needs to be further explored.

In a multicenter survey conducted in China, preoperative anxiety was found to be related to more severe pain postoperatively.³⁹ In this study, EA showed a better effect in improving the 24 hours postoperative VAS score. Preemptive analgesia, also known as protective analgesia, refers to the active start of multimodal intervention before surgery, and its continuation through the entire perioperative and

postoperative periods. This approach has been shown to reduce postoperative pain.⁴⁰ The preoperative intervention of acupuncture is also in line with this concept and can play a role in preemptive analgesia.

In this study, intraoperative EA improved the physical comfort score and the quality of life of the patients, while combination of preoperative and intraoperative EA not only most significantly improved the quality of life at 72 hours postoperatively, but markedly improved the QoR-40 total score and physical comfort. Moreover, factorial analysis revealed an interactive effect between them, indicating that concomitant use of preoperative EA and intraoperative EA may lead to best improvement in scores.

Apart from this, our study also demonstrated that the more severe was the preoperative anxiety state, the worse was the quality of the postoperative recovery. Therefore, starting the intervention 1 day before surgery can not only optimize the preoperative state of the patient, and alleviate stress during the operation, but also improve the postoperative quality of life of the patient. The combined technique acupuncture group had the greatest clinical significance and is worth recommending.

A limitation of this study was that we did not include a sham acupuncture group. In general, the main types of auxiliary tools used for sham acupuncture are blunt needles, ordinary acupuncture needles, TEAS instruments, or other non-acupuncture placebo needles. The effect of sham acupuncture may include the specific and nonspecific effects of acupuncture in addition to its placebo effect which may interfere with the results.⁴¹ Another limitation of this study was that it was a single-center study. Further multi-center studies with larger sample size are required to provide more robust evidence.

One of the strengths of this study is its analysis method. By using factorial design, we were able to assess the effect of each group on the outcome variables, as well as the effects of interactions between different groups, particularly between the preoperative and intraoperative acupuncture. This analytic approach can allow the physicians to choose the best technique for obtaining optimal results.

Conclusions

In this study, EA alleviated patient anxiety in the preoperative waiting area and improved sleep quality, reduced the incidence of PONV, relieved postoperative pain, and improved the QoR-40 total score and emotional state of patients at 72 hours postoperatively. Compared with preoperative EA and intraoperative EA, combined preoperative and intraoperative EA had better efficacy. No serious adverse reactions occurred in this study, indicating the safety of acupuncture. Electroacupuncture may be a useful intervention as part of the ERAS approach for breast cancer; however, large-scale multi-center clinical trials are required to validate our findings.

Clinical Practice Points

Patients with breast cancer may have a higher risk of preoperative anxiety. The traditional method of anxiolytic medication often cause serious side effects. Postoperative opioid use tends to affect outcomes, while psychological nursing also has certain limitations. ERAS protocols for perioperative care can lead to improvement in clinical outcomes, and particularly ameliorate preoperative

anxiety. Acupuncture therapy has been used in ERAS pathways to reduce perioperative pain and postoperative nausea and vomiting. Few studies have shown that it can reduce preoperative anxiety, but it was mainly conducted in the context of neuro-, cataract, and gynecological surgeries. The findings of the present study showed that preoperative EA alleviated anxiety in the preoperative waiting area, and improved sleep quality, while combination of preoperative and intraoperative EA may be more effectively improve postoperative quality of life. EA could be a useful intervention as part of the ERAS approach for breast cancer.

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Disclosure

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Author Contributions

QYT and RL designed the study and wrote and revised the draft manuscript and subsequent manuscripts. WDS, RL, and QYT revised the draft manuscript and subsequent manuscripts. QYT, YG, and KZ assisted with the presentation of findings and assisted with drafting and revising the manuscript. WM and WDS conceived and designed the study, assisted with literature search, appraised relevant studies, and assisted with drafting and revising the manuscript. All authors read and approved the final manuscript.

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